TIDJANI BACHIR

Full-Stack Developer

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GitHub: github.com/Tidjani1Bachir

Portfolio: personnel-website-dusky.vercel.app

SUMMARY

Full-Stack Developer with hands-on experience building MERN stack applications, Next.js platforms, and TypeScript-based tools. Passionate about clean architecture, security-first design, and delivering production-ready applications. Currently pursuing a Master's in Software Engineering.

PROJECTS

Full-Stack E-Commerce Platform — MERN Stack:

Architecture & Security: Built with JWT Bearer token authentication, bcrypt password hashing, 4-layer file upload validation (extension, MIME type, path traversal, malware scanning), and NoSQL injection protection via express-mongo-sanitize with explicit type casting across all controllers

- **State Management:** RTK Query tag-based cache invalidation for real-time UI sync on all admin CRUD operations; hybrid localStorage/MongoDB favorites for guest-to-user data migration
- **Payment & Dashboard:** Integrated PayPal SDK for PCI-compliant payments; admin dashboard with live ApexCharts analytics and CSV/PNG/SVG exportable reports
- **Deployment:** Decoupled architecture — React on Vercel, Node.js/Express API on Render, MongoDB Atlas cloud database with automated seeding scripts

Product Manager — Web + Desktop App:

- **Live App:** https://product-manager-chi-eosin.vercel.app
- **Desktop (.exe):** https://github.com/Tidjani1Bachir/product_manager/releases/download/v2.0-desktop/product-manager_0.1.0_x64-setup.exe
- **Desktop (.msi):** https://github.com/Tidjani1Bachir/product_manager/releases/download/v2.0-desktop/product-manager_0.1.0_x64_en-US.msi
- **Stack:** React, TypeScript, Tauri, Node.js/Express with full CRUD, drag-and-drop image upload, and PDF export
- **Deployment:** Dual-version architecture — web app on Vercel (serverless) and native desktop installer (.exe / .msi) via Tauri from a single GitHub repository
- **Cloud:** Integrated Turso (cloud SQLite), Cloudinary (image storage), and RESTful Express API for a zero-infrastructure-cost cloud-native stack
- **Testing:** 39/39 passing tests — Vitest + React Testing Library (unit), Playwright E2E across Chrome, Firefox & Safari, and node:test for REST API validation
- **TypeScript Migration:** Migrated full codebase from JavaScript to TypeScript with strict typing, shared interfaces, and a centralized API service layer

Shoppy — React Admin Dashboard:

Live App: https://react-admin-dashboard-rbrf.vercel.app

- **Feature-Rich Interface:** Built an admin dashboard featuring Syncfusion EJ2 charts, data grids, and Kanban boards for complex data visualization.
- **Dynamic Customization:** Integrated dark/light theming with six live accent colors via React Context API and Tailwind CSS.

- **Modular Architecture:** Engineered reusable UI components and managed complex application state for a seamless user experience.
- **Optimized Deployment:** Deployed the production-ready application on Vercel to ensure high performance and global availability.

Brainwave AI — Modern SaaS UI/UX

- **Live App:** https://modern-saas-ptototype.vercel.app/
- • **Figma-to-Code Mastery:** Developed a high-fidelity React implementation of a premium AI SaaS design, ensuring 1:1 pixel-perfect accuracy.
- • **Advanced UI/UX:** Engineered complex dark-mode layouts and smooth animations using Tailwind CSS and CSS pseudo-elements.
- • **Responsive Design:** Optimized the architecture for flawless responsiveness and interactive states across all device types.
- • **Performance:** Leveraged Vite for near-instant load times and high-performance rendering of visual-heavy assets.

EDUCATION

Master of Science in Software Engineering (In Progress) 2024 – Present

University of Saad Dahleb – Blida, Algeria

- Relevant coursework: Software Architecture, Advanced Development Methodologies, Project Design

Bachelor of Science in Computer Systems 2020 – 2024

University of Saad Dahleb – Blida, Algeria

- Final Year Project: Lawyer-client management platform using Firebase (backend) and Android Studio / Java (mobile)
- Features: real-time database sync, user authentication, appointment & case management

Scientific Baccalaureate — Bien (Above Average) 2021

Houari Mahfoud High School – Blida, Algeria

- Specialization: Experimental Sciences

TECHNICAL SKILLS

Frontend: React, Next.js 14, TypeScript, Tailwind CSS, Redux Toolkit / RTK Query, Headless UI, Framer Motion, Syncfusion EJ2

Backend: Node.js, Express.js, REST API Design, JWT Authentication, bcrypt, Multer, Rate Limiting, File Upload Validation, express-mongo-sanitize

Database: MongoDB Atlas, MongoDB Compass, Turso (cloud SQLite), Firebase

DevOps: Vercel, Render, GitHub Pages, GitHub Actions, Tauri (desktop), Cloudinary

Testing: Vitest, React Testing Library, Playwright (E2E), node:test, Postman (API testing)

Tools: Git & GitHub, Figma (reading designs), Slack, Jira, ApexCharts, PayPal SDK, RapidAPI

OS & Scripting: Debian-based Linux — command-line proficiency, package management (apt/dpkg) for installing software such as MongoDB, Postman, and language runtimes; file system navigation, directory & file management, permissions & ownership (chmod, chown, chgrp); text editing with Vim & Nano; setting up virtual machines with Linux distributions (VirtualBox / VMware); working with WSL (Windows Subsystem for Linux); Windows.

Shell Scripting: Bash scripting including automated website uptime monitoring — script polls a URL every minute and sends an email alert if the site is unresponsive; general-purpose automation scripts for system tasks and workflows

Local AI: Setting up local AI environments using Ollama for private, offline inference and AI-assisted development workflows

Other: UML Diagrams (Use Case, Class, Activity, State Machine, Sequence), Agile, Clean Architecture, MVC Pattern

Languages: Arabic (Native), English (Intermediate), French (Intermediate)